LESSON PLAN

GUIDED WEAPON (37)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Bomb ballistics – behaviour of bomb when released from aircraft in wind condition
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Bomb ballistics –behaviour of bomb when released from aircraft in wind condition
5. Specific Objectives	37.1.1.1 Trainee will be able to explain about various branches of ballistics

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Trajectory Charts; Instructor supervision required for chart interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three main branches of ballistics?
(b) Define the term 'ballistic coefficient'.
(c) List two factors affecting bomb trajectory.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to Ballistics • External Ballistics Definition • Internal Ballistics Overview • Terminal Ballistics Overview • Relevance to Bombing	L & Ds	PPT, IB	15 min
(b)	• Effect of Wind on Bomb • Wind Drift Calculation • Crosswind Component • Headwind and Tailwind • Trajectory Correction	L & Ds	PPT, PDF	20 min
(c)	• Bomb Release Mechanics • Airspeed and Altitude • Glide Angle Analysis • Impact Point Prediction • Summary of Ballistic Factors	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how crosswind affects the bomb's impact point.
(b) Differentiate between internal and external ballistics in the context of bomb release.

12. Home Assignment and Follow Up	Prepare a brief note on how wind conditions influence the release point calculation for the next lesson on bombing procedures.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min